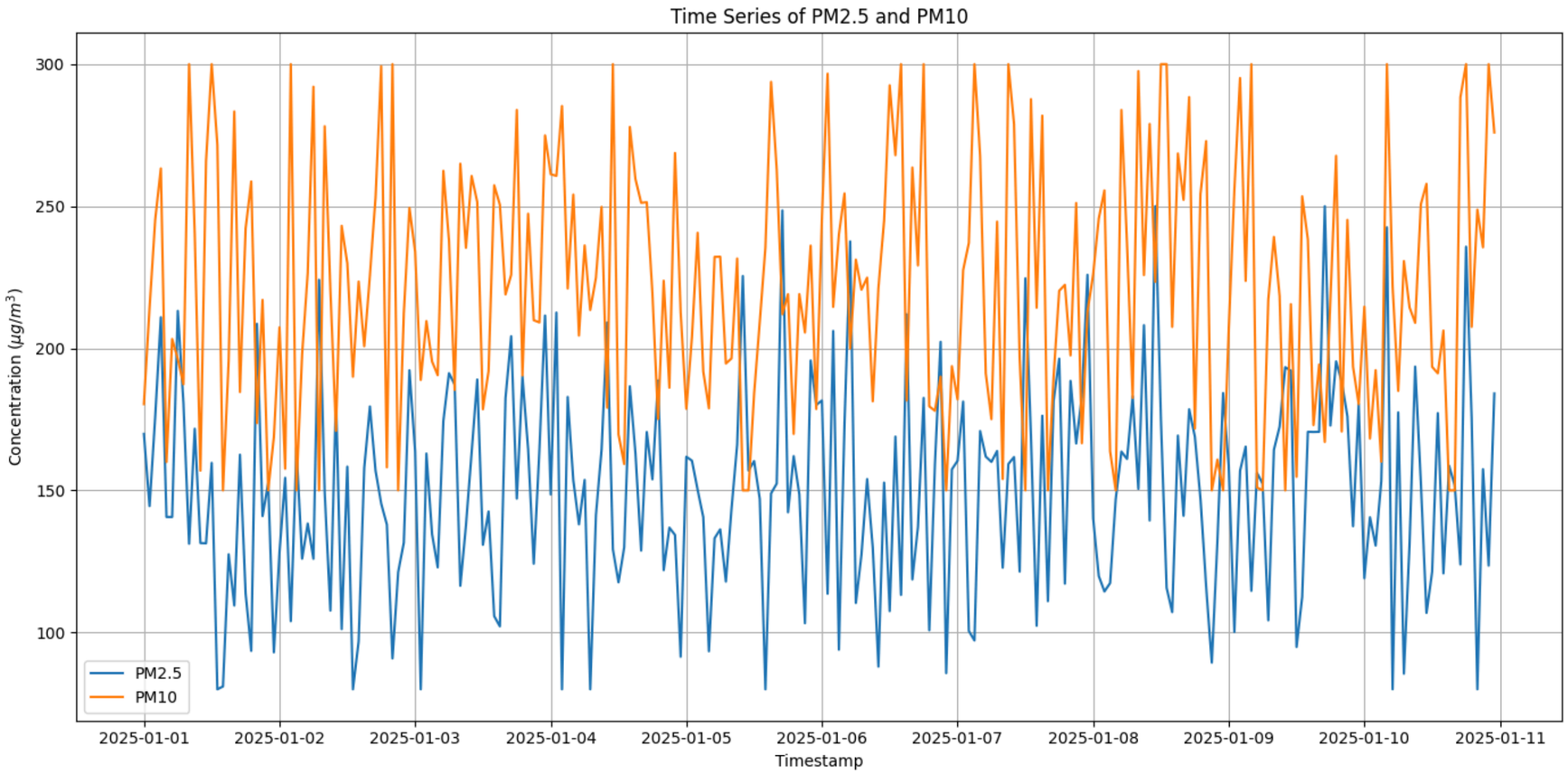
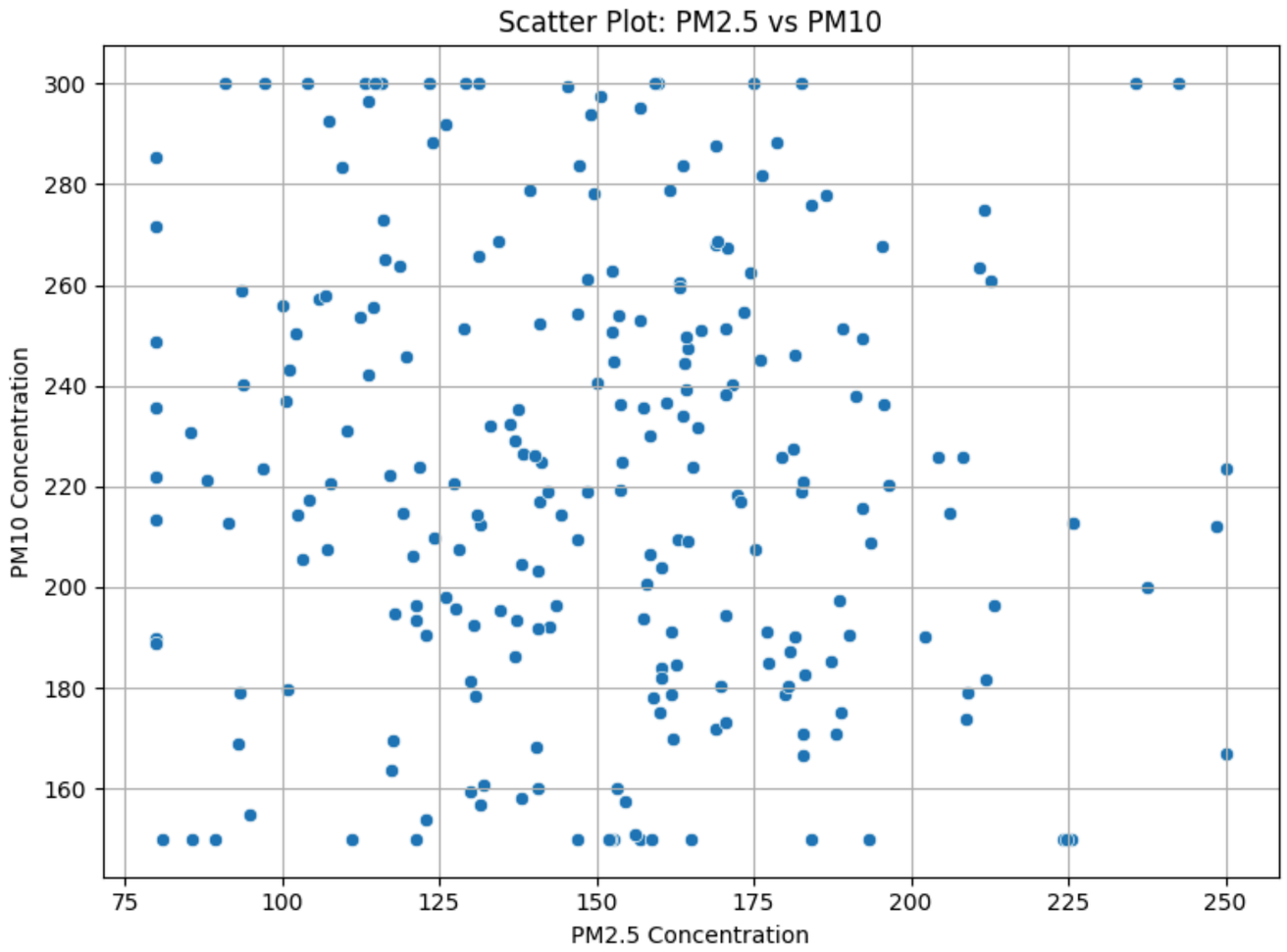
# Conduct EDA to understand pollutant trends and correlations



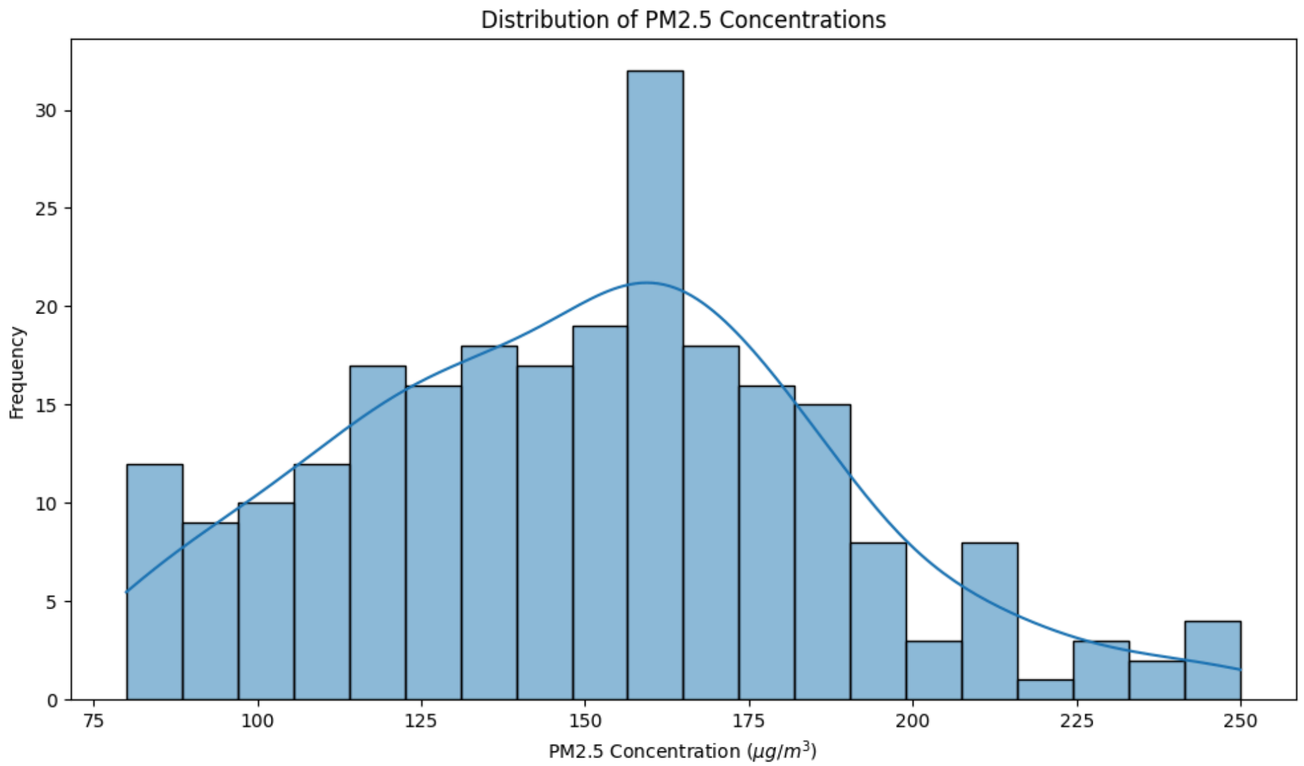

**Distribution of PM2.5 Concentrations**

- **What it shows:** This is a histogram that illustrates the frequency of different PM2.5 concentration levels. The x-axis shows concentration ranges, and the y-axis shows how many data points fall into each range.
- **What we learn:**
  - **Common Range:** The tallest bar indicates the most common range of PM2.5 concentrations.
  - **Skewness:** The distribution is likely skewed to the right, with a long tail extending toward higher concentration values. This is common with environmental data, showing that most of the time the pollution is at a moderate level, but there are occasional, less frequent spikes.
  - **Potential Outliers:** The very long tail in the right side indicates that there are some very high concentration values, confirming the presence of potential outliers.



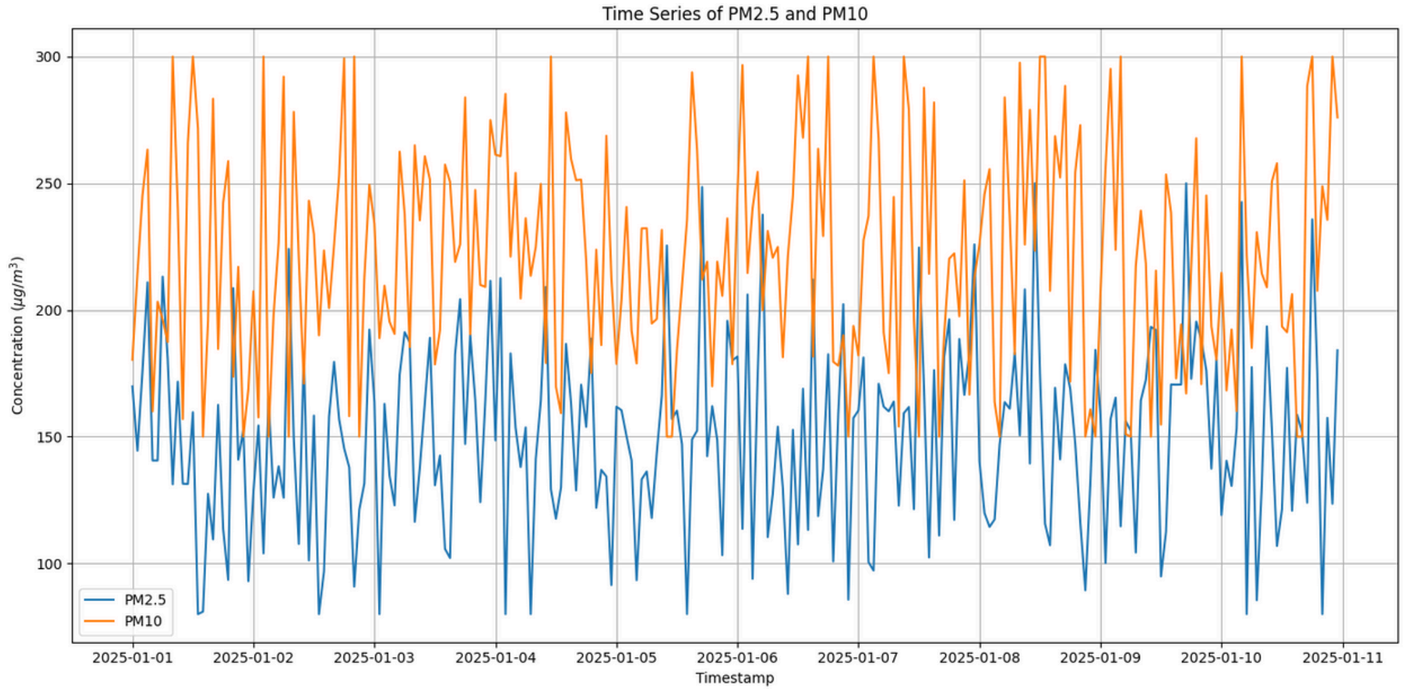

**Scatter Plot of PM2.5 vs PM10**

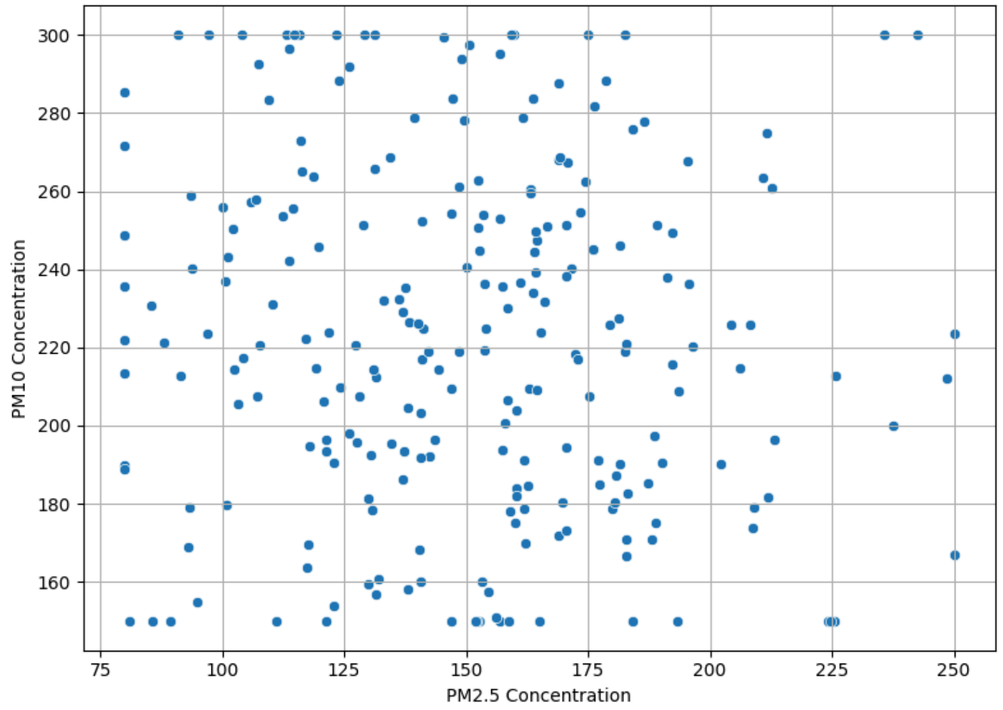
- **What it shows:** This plot visualizes the relationship between the PM2.5 and PM10 concentrations. Each dot represents a single data point (an hour) with its corresponding PM2.5 and PM10 values.
- **What we learn:**
  - **Strong Correlation:** The data points form a clear linear pattern, indicating a strong positive correlation between PM2.5 and PM10. This means as PM2.5 levels increase, PM10 levels also tend to increase.
  - **Outliers:** You might spot a few points that deviate from the main cluster. These could be potential outliers that need to be investigated.

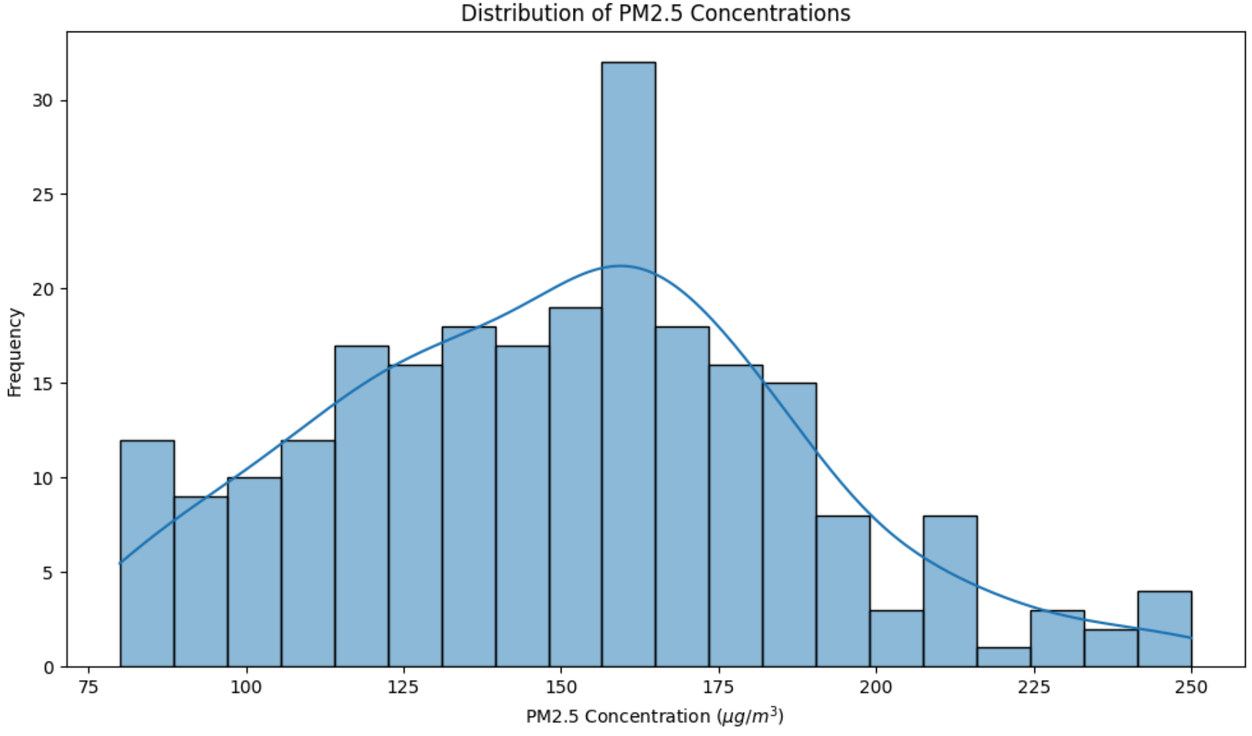



**Time Series of PM2.5 and PM10**

- **What it shows:** This plot displays the concentrations of PM2.5 and PM10 over time. The x-axis represents the timestamp, and the y-axis shows the pollutant concentration.
- **What we learn:**
  - **Trends:** We can see the overall trend of both pollutants over the recorded period.
  - **Volatility:** The jagged lines indicate significant hourly fluctuations in air quality.
  - **Relationship:** The PM10 concentration is consistently higher than PM2.5, which is expected since PM10 includes PM2.5 particles. The two pollutants also appear to move together, suggesting that factors affecting one, like traffic or industrial emissions, also affect the other.


Time Series of PM2.5 and PM10
Concentration ($\mu g/m^3$)
300
250
200
150
100
PM2.5
PM10
2025-01-01
2025-01-02
2025-01-03
2025-01-04
2025-01-05
2025-01-06
2025-01-07
2025-01-08
2025-01-09
2025-01-10
2025-01-11
Timestamp


PM10 Concentration
300
280
260
240
220
200
180
160
75
100
125
150
175
200
225
250
PM2.5 Concentration


Distribution of PM2.5 Concentrations
Frequency
30
25
20
15
10
5
0
75
100
125
150
175
200
225
250
PM2.5 Concentration ($\mu g/m^3$)

# Statistical Summary of Pollutant Data

```
(venv) PS C:\Users\ktumm\OneDrive\Desktop\AirQualityProject> python -u "c:\Users\
e.py"
count  240.000000  240.000000  240.000000  240.000000  240.000000  240.000000
mean   149.984168  222.002561   18.625957   69.360464    0.843438   60.872938
std     37.254414   44.009045    4.650204   18.772760    0.281655   19.025243
min     80.000000  150.000000   10.000000   40.000000    0.400000   20.000000
25%    122.578727  188.482153   15.218308   54.838443    0.628457   46.832081
50%    152.490205  220.570637   18.871416   68.769746    0.838052   60.660876
75%    173.722476  254.148880   21.407992   81.926569    1.039853   72.305956
max    250.000000  300.000000   30.000000  120.000000    1.500000  100.000000
```

- **Top-Left Metrics:** The cards for MAE, RMSE, and R-squared provide a quantitative summary of the model's accuracy. A lower MAE and RMSE indicate smaller prediction errors, while an R-squared closer to 1 shows a better fit to the data.
- **Main Plot (Actual vs. Forecasted):** This chart shows the model's predictions (orange line) against the real historical data (blue line). This is the most crucial part of the dashboard as it lets you visually evaluate how well the model works.
- **Bottom-Right Plot (24-Hour Forecast):** This plot provides the final, actionable prediction by showing the model's forecast for the next 24 hours.